[Exercise 1 - Filtering Statements]

1. SELECT *
FROM employees_db.schema.employees;

RESULT:

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT department
FROM employees_db.schema.employees;

RESULT:

department
IT
HR
Finance
Marketing

3. SELECT first_name,
last_name,
salary
FROM employees_db.schema.employees
ORDER BY salary DESC;

(Question 3)

RESULT:

first_name	last_name	salary
Emily	Davis	62000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53000
Daniel	Clark	53 000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50 000
Jane	Smith	48000

4.
```
SELECT first_name,
       last_name,
       salary
FROM employees_db.schema.employees
ORDER BY salary DESC
LIMIT 5;
```

RESULT:

first_name	last_name	salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55 000
Sarah	Brown	53 000

5.
```
SELECT first_name,
       last_name,
       department
FROM employees_db.schema.employees
WHERE department = 'IT';
```

(Question 5)

RESULT:

first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6.
```
SELECT first_name,
       last_name,
       department,
       salary
FROM employees_db.schema.employees
WHERE department = 'Finance'
AND salary > 58000;
```

RESULT:

first_name	last_name	department	salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000

7.
```
SELECT first_name,
       last_name,
       department
FROM employees_db.schema.employees
WHERE department = 'HR'
OR department = 'Marketing'
```

RESULT:

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

8. SELECT first_name,
last_name,
department
FROM employees_db.schema.employees
WHERE department != 'IT';

RESULT:

first_name	last_name	department
Jane	Smith	HR
Mike	Johson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. SELECT first_name,
last_name,
department
FROM employees_db.schema.employees
WHERE department IN ('HR', 'IT', 'finance');

RESULT:

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10. SELECT first_name,
last_name,
department,
salary,
city
FROM employees_db.schema.employees
WHERE department = 'IT'
AND salary > 50000
AND city = 'New York';

RESULT:

first_name	last_name	department	city	salary
John	Doe	IT	New York	55 000
Sarah	Brown	IT	New York	53 000

11. SELECT first_name,
last_name,
department,
salary
FROM employees_db.schema.employees
WHERE department = 'Finance'
OR department = 'Marketing'
AND salary > 52000
ORDER BY salary DESC;

RESULT:

first_name	last_name	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	5300

12. SELECT DISTINCT city
FROM employees_db.schema.employees
WHERE department != 'IT'
OR department != 'HR';

RESULT:

city
Los Angeles
San Francisco
Houston
Chicago

13. SELECT first_name,
last_name,
department,
salary,
hire_date
FROM employees_db.schema.employees
WHERE department != 'Finance'
AND salary > 50000
ORDER BY hire_date ASC;

RESULTS:

first_name	last_name	department	salary	hire_date
Emily	Davis	IT	62000	2015-02-14
David	White	Marketing	52000	2016-04-10
Jessica	Moore	HR	51000	2018-05-22
John	Doe	IT	55000	2018-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

14. SELECT first_name,
last_name,
department,
city
FROM employees_db.schema.employees
WHERE department = 'IT'
OR department = 'Marketing'
AND city = 'Chicago'
OR city = 'Los Angeles'
LIMIT 3;

RESULT:

first_name	last_name	department	city
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago
NULL	NULL	NULL	NULL